A WORKBENCH FOR FREECAD

# InCad

A familiar Autodesk Inventor and Alibre style interface on top of the FreeCAD kernel

Test build 0.3 · macOS on Apple Silicon · FreeCAD 1.1

WHY

# FreeCAD's kernel is strong. Its interface feels foreign if you come from Inventor

## A new interface

Ribbon, model browser, dialogs and menus as in Inventor and Alibre

## The same kernel

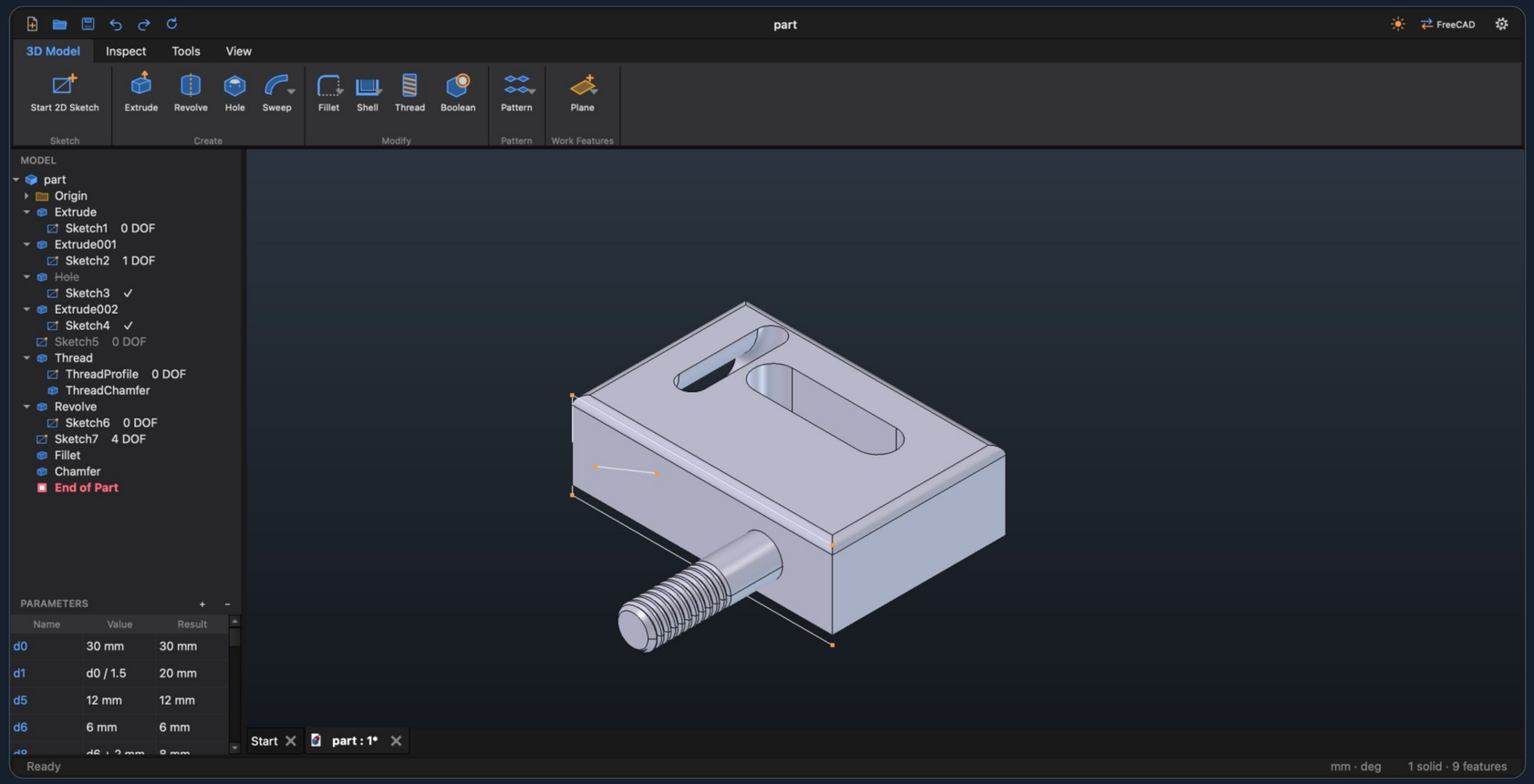
Under the hood: FreeCAD's own Sketcher and PartDesign

## Ordinary files

Parts are saved as .FCStd and open in any FreeCAD

A FAMILIAR LAYOUT

# This is how it looks




The ribbon on top, the model browser and parameters on the left, the model in the middle. Navigation on the right.

InCad with a real part: a block with a threaded shaft, slots and rounds.

WHAT'S INSIDE

# Four pillars of the interface

## Command ribbon

3D Model, Inspect, Tools and View tabs, and tabs of its own in a sketch

## Model browser

One history per part, sketches under their features, End of Part

## Dialogs with preview

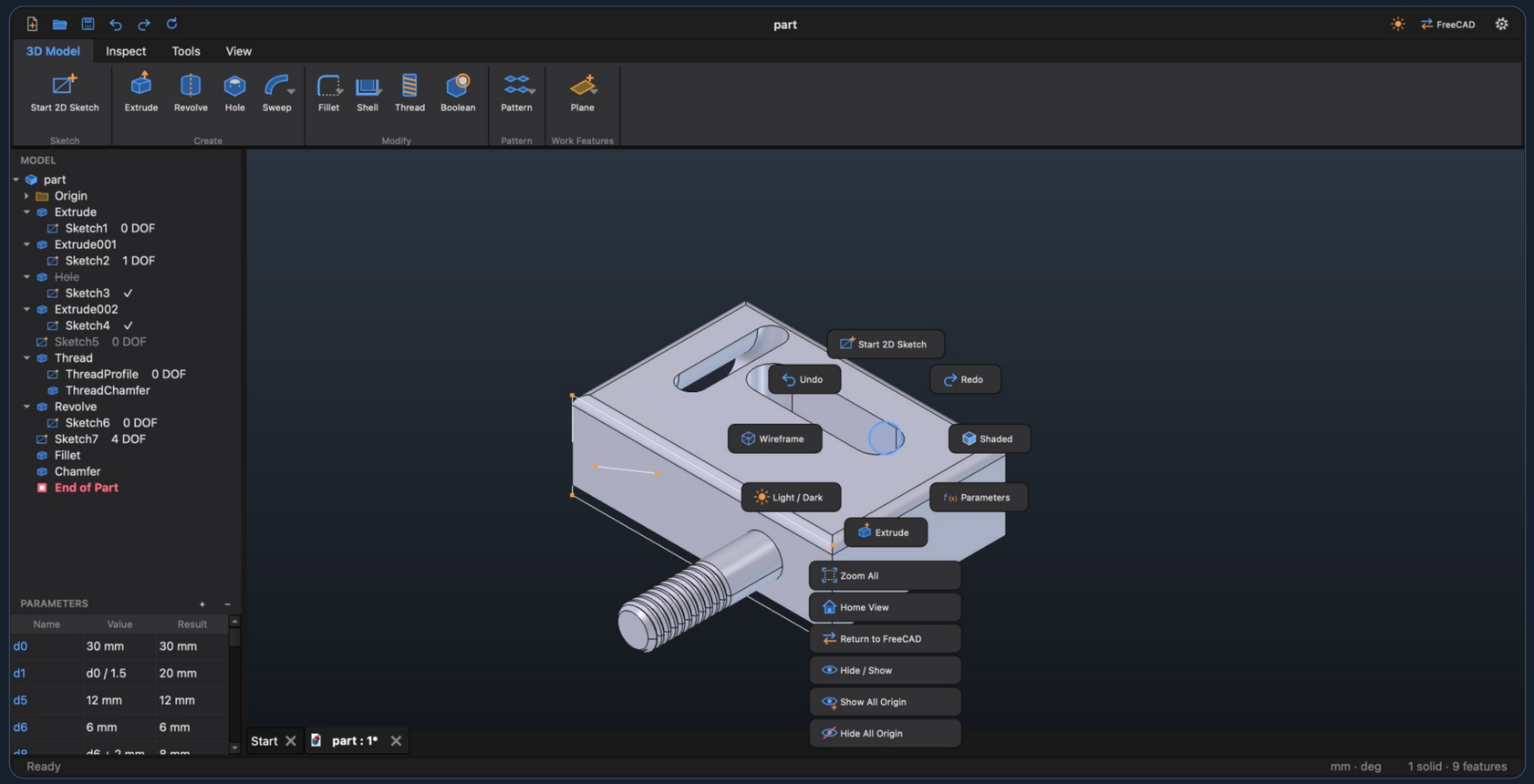
See the result at once, ▲▼ step the numbers, Undo inside the dialog

## Marking menus

The right button opens a radial menu, a flick of the mouse picks the command

RIGHT MOUSE BUTTON

# Marking menus

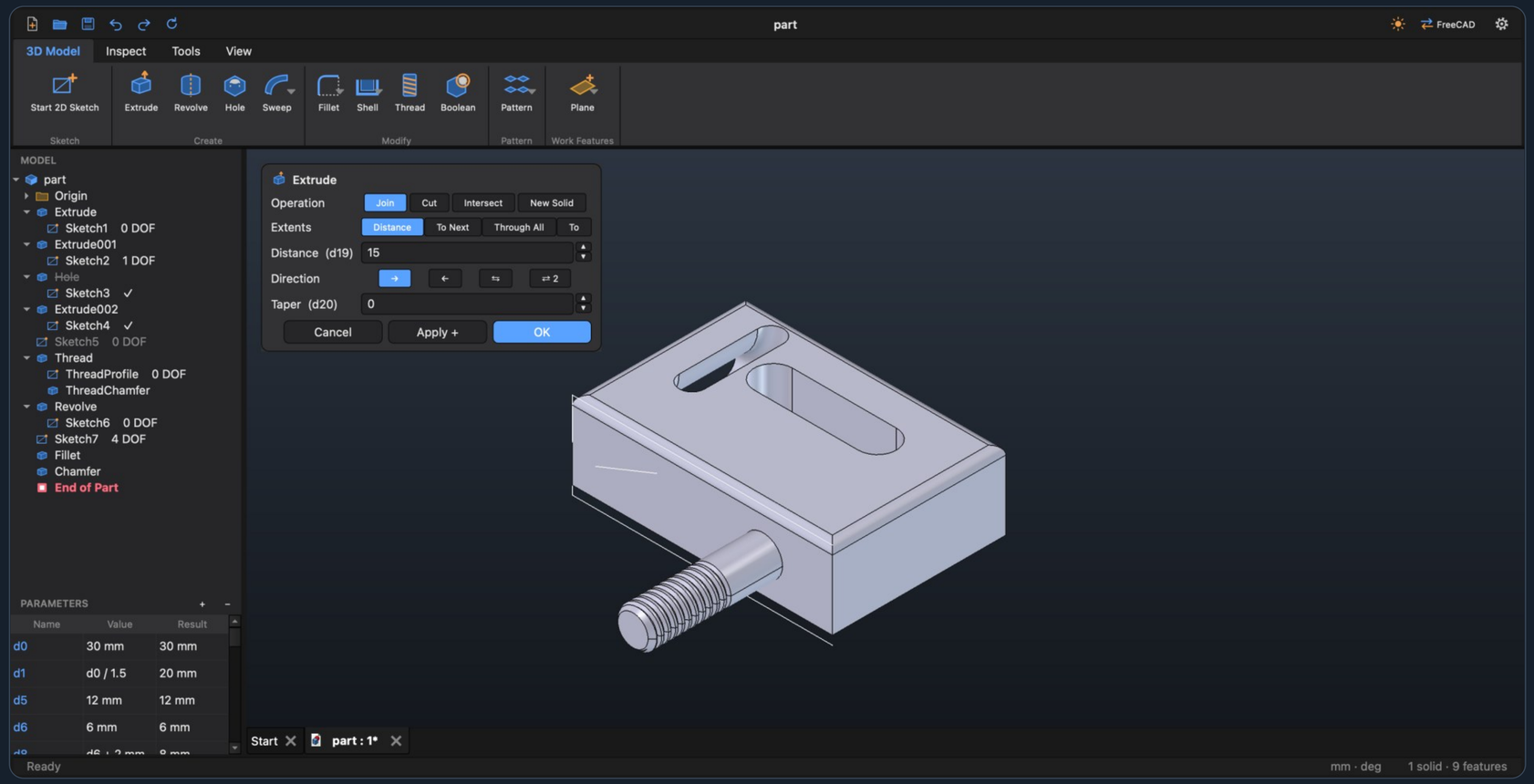



Commands around the ring are picked with a flick of the mouse, frequent actions are listed below.

Near a screen edge the menu moves inward; the gesture still starts where you pressed.

EDITING AN EXTRUDE

# Feature dialog




Every change shows on the model at once. The ▲▼ arrows step the number, Shift steps by 10.

The (d19) label means the value is a model parameter you can use in formulas.

MODEL BROWSER

# One history, as in Inventor

- **Every solid in one list** in the order made, rows of other solids name theirs
- **One End of Part**: drag the marker and the whole part rolls back to that point
- **Highlight both ways**: a feature in the browser shows its faces, a face in the view shows its feature

IN THE MENU

### Show in Browser

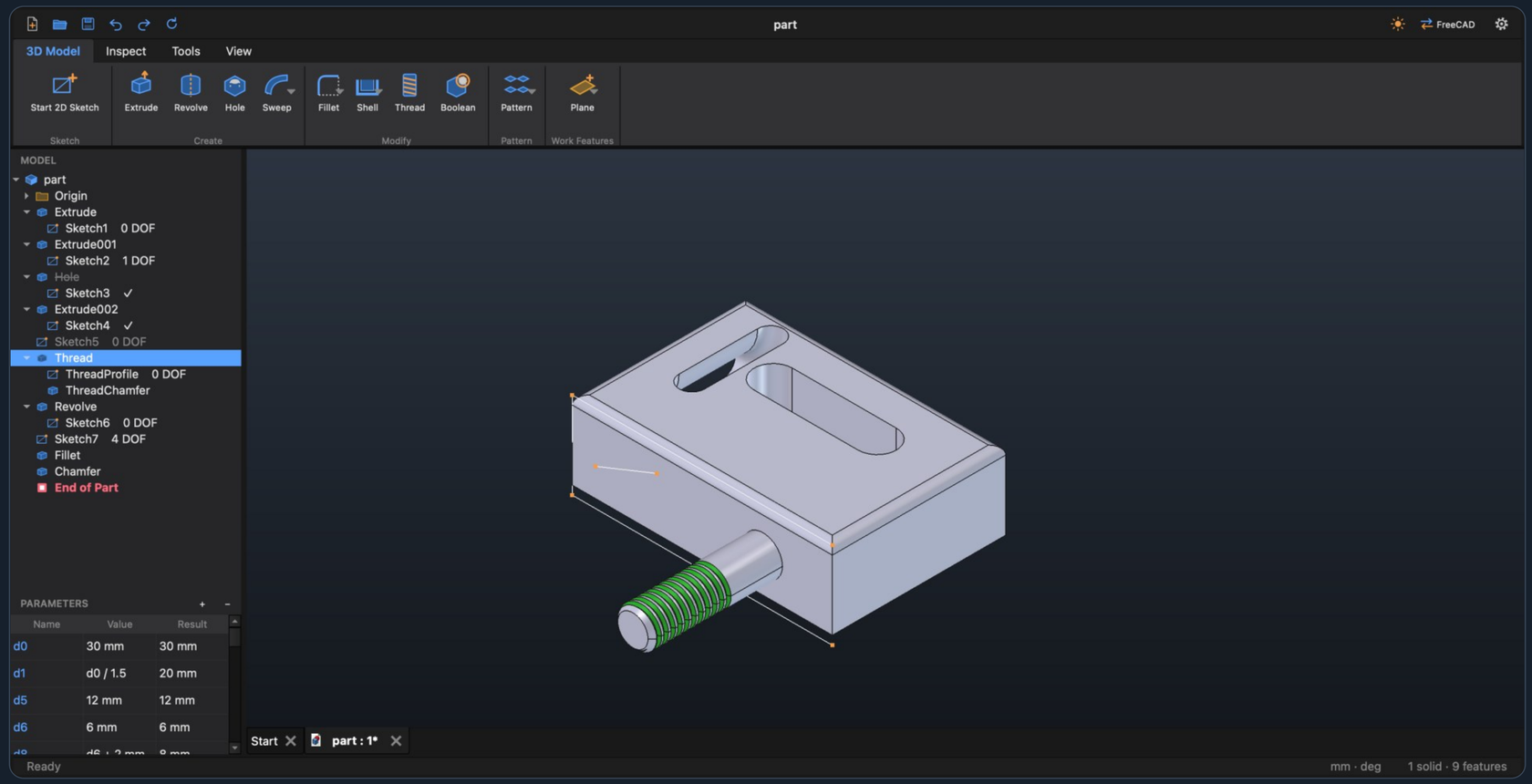
from the view: open the browser at the feature of the picked face

### Show in Window

from the browser: light the feature's faces and zoom to them

BROWSER ↔ VIEW

# Feature highlight




The Thread row in the browser lights the faces the thread made on the model.

It works the other way too: a click on a face lights its feature in the browser.

SKETCHES AND PARAMETERS

# Named dimensions and formulas

## Parameters

Each dimension gets a name, d0, d1… Write d0 / 1.5 and change everything from the Parameters panel

## Patterns in a sketch

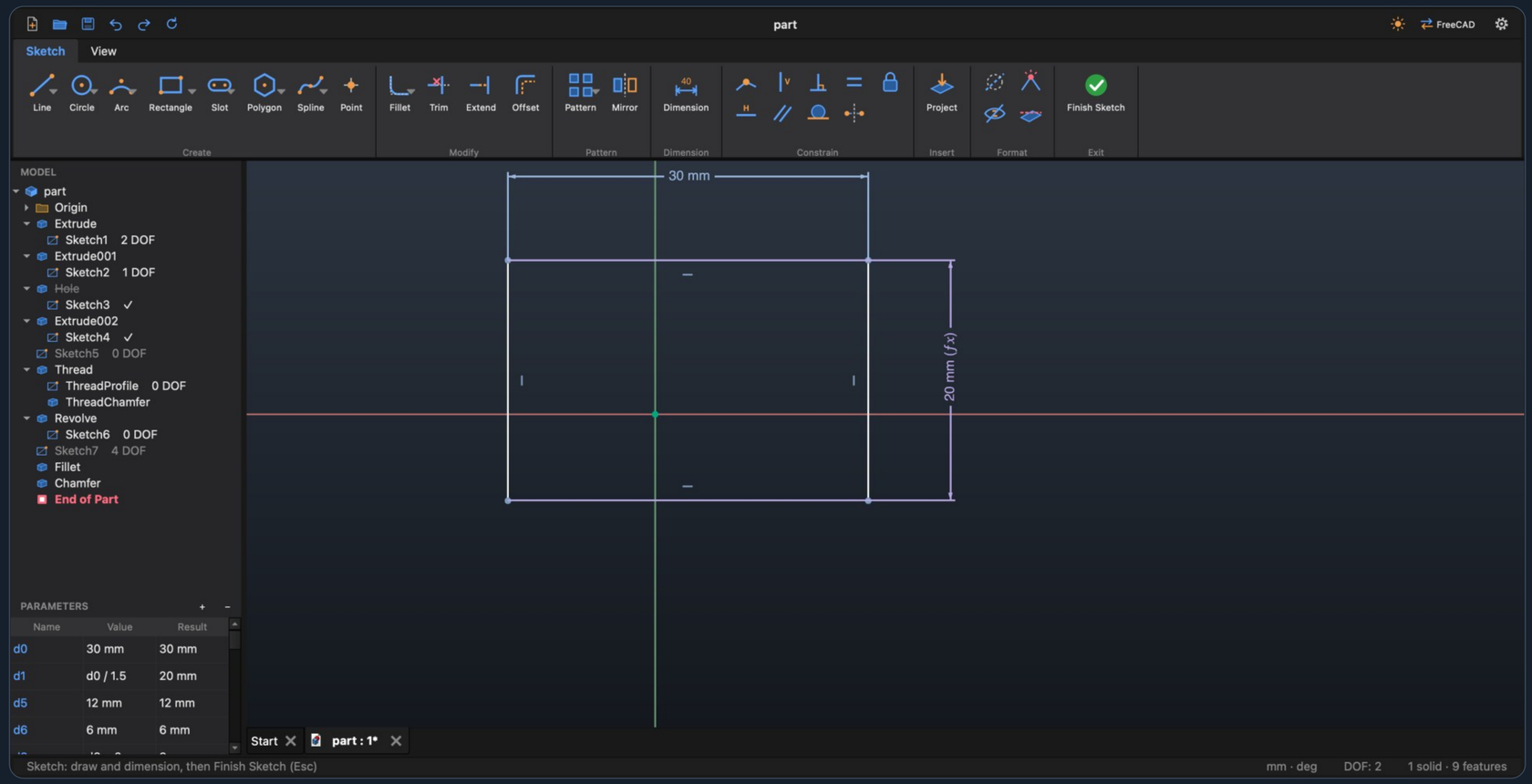
Rectangular, circular and along a path, with equal spacing between limits

## Open contours

Paths for Sweep and Loft stay visible with their ends and points

A RIBBON OF ITS OWN FOR SKETCHING

# Sketch mode




Drawing, editing, patterns, dimensions and constraints sit on one Sketch tab.

The 20 mm dimension is a formula (fx): d0 / 1.5 from the Parameters panel.

FEATURES

# Everything for a part in one ribbon

| Group | Commands |
|---|---|
| Create | Extrude, Revolve, Hole (with Chamfer), Sweep, Loft, Coil |
| Modify | Fillet, Chamfer, Shell, Thread, Draft, Boolean (Keep tool), Split, Threads Off / On |
| Patterns | Rectangular and Circular Pattern, Mirror |
| Work features | Plane, Axis, Point |
| Assemble | Make Assembly |

**Sweep with a wall thickness:** pipes and ducts in one feature, one end or both open

NO SURPRISES

# InCad takes care of what trips you up in FreeCAD

## Separate solids

A box drawn away from the part becomes a solid of its own

## The right solid

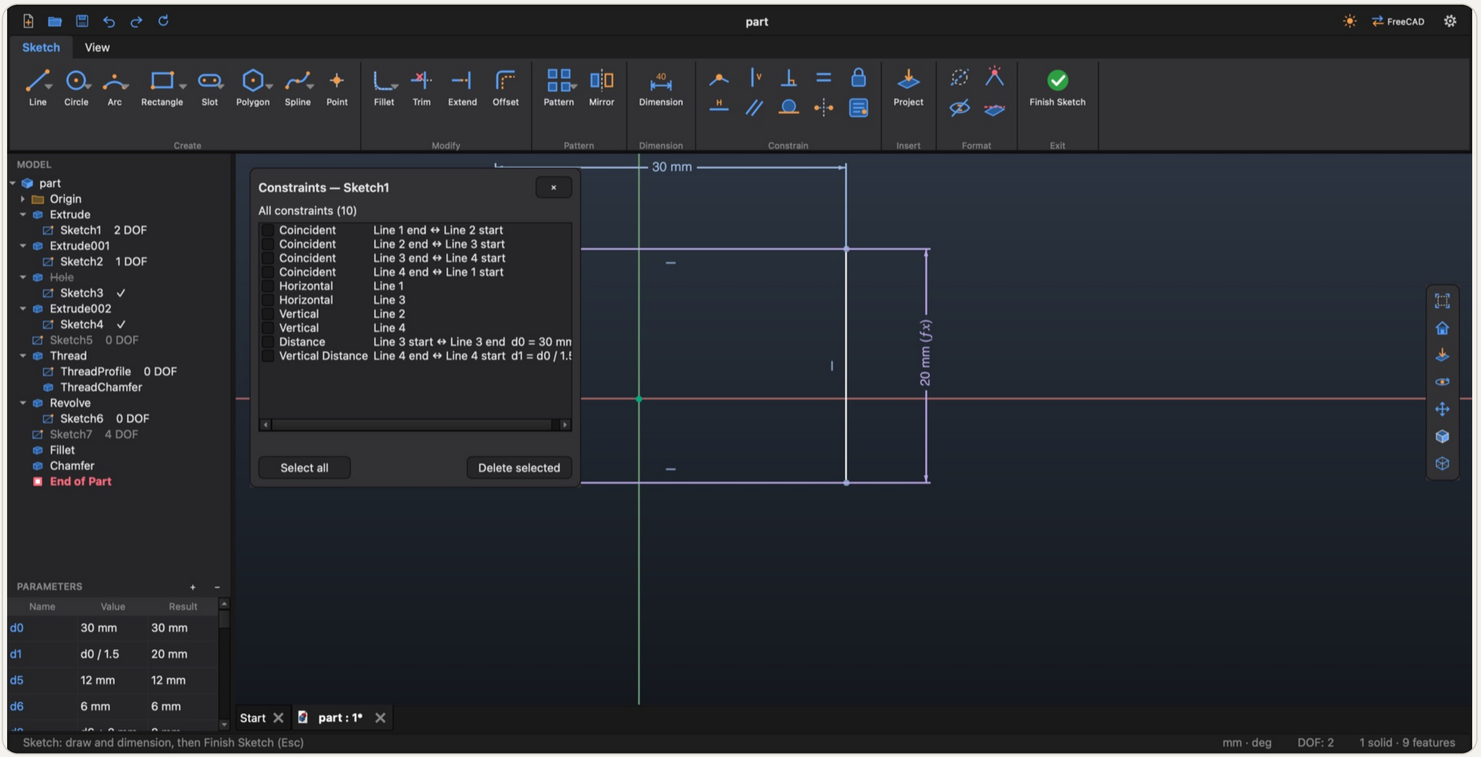
A sketch on a solid's face, and a Cut made from it, go to that solid

## No freezes

A risky Loft is tried on a copy first instead of freezing FreeCAD

SKETCH

# Constraint List




Every constraint of the sketch in one list, also the ones with no icon to click.

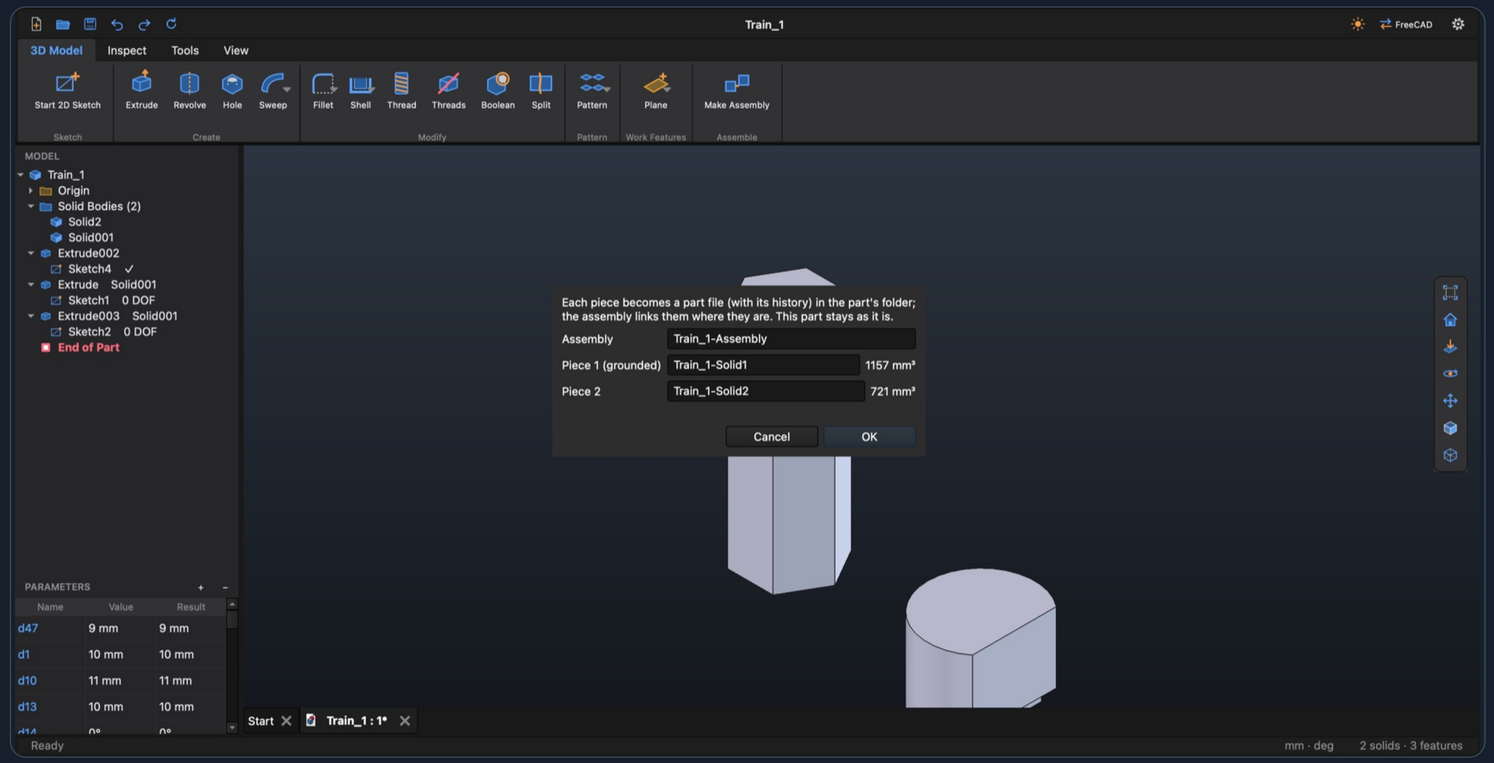
Pick a line or point to see only its own. Check rows and press Delete selected.

Or right-click a line or point and delete one of its constraints.

Projected edges are normal lines by default; Construction switches them.

NEW IN PARTS

# Split and Make Assembly




## SPLIT

The lines of a sketch cut the part. Keep all: each piece is a Solid. Keep picked: only the pieces you click stay.

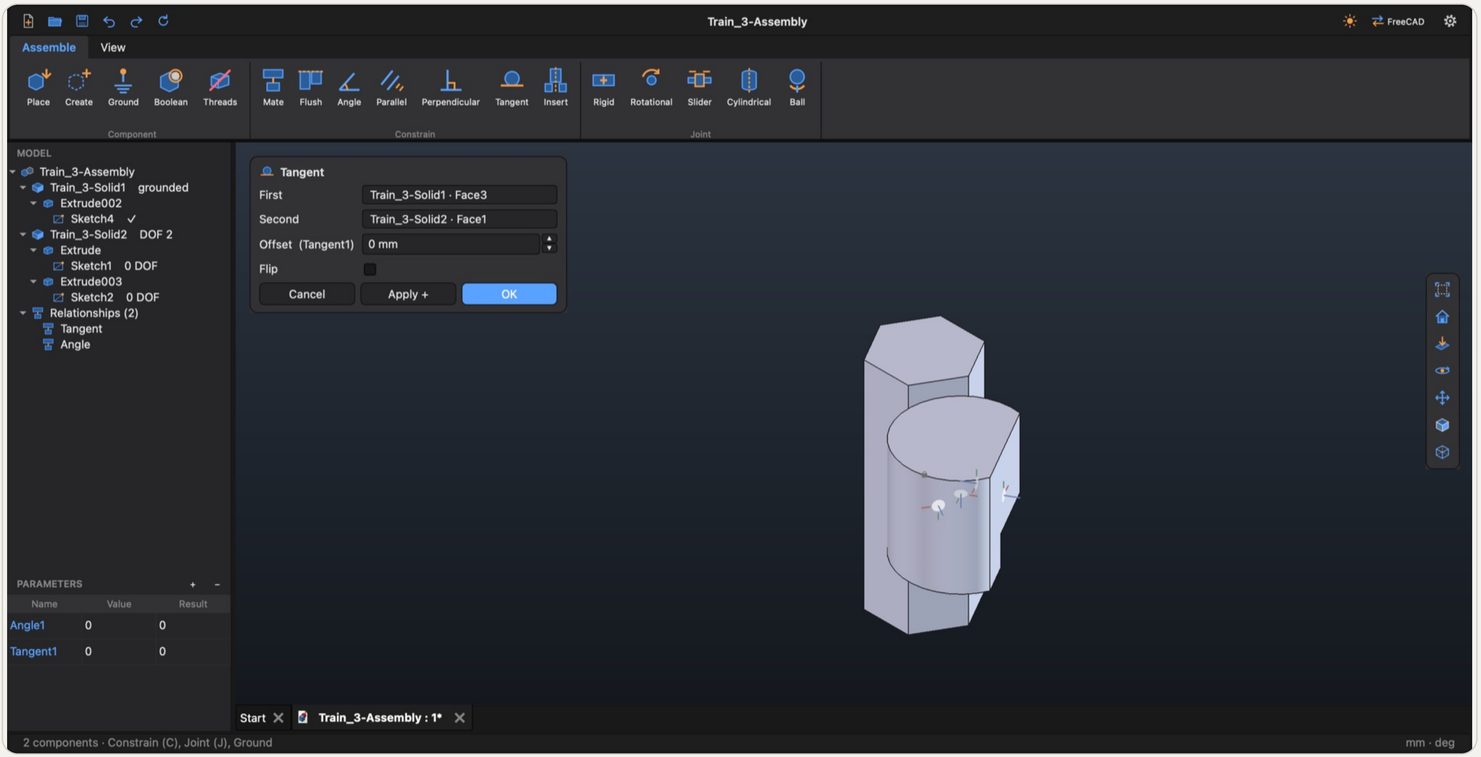
## MAKE ASSEMBLY

Each separate piece becomes a part file with its history; a new assembly links them where they were.

Run it again and choose Replace existing files: the old files are overwritten.

ASSEMBLIES

# Constraints and joints




## CONSTRAINTS

Mate, Flush, Angle, Parallel, Perpendicular, Tangent, Insert

## JOINTS

Rigid, Rotational, Slider, Cylindrical, Ball

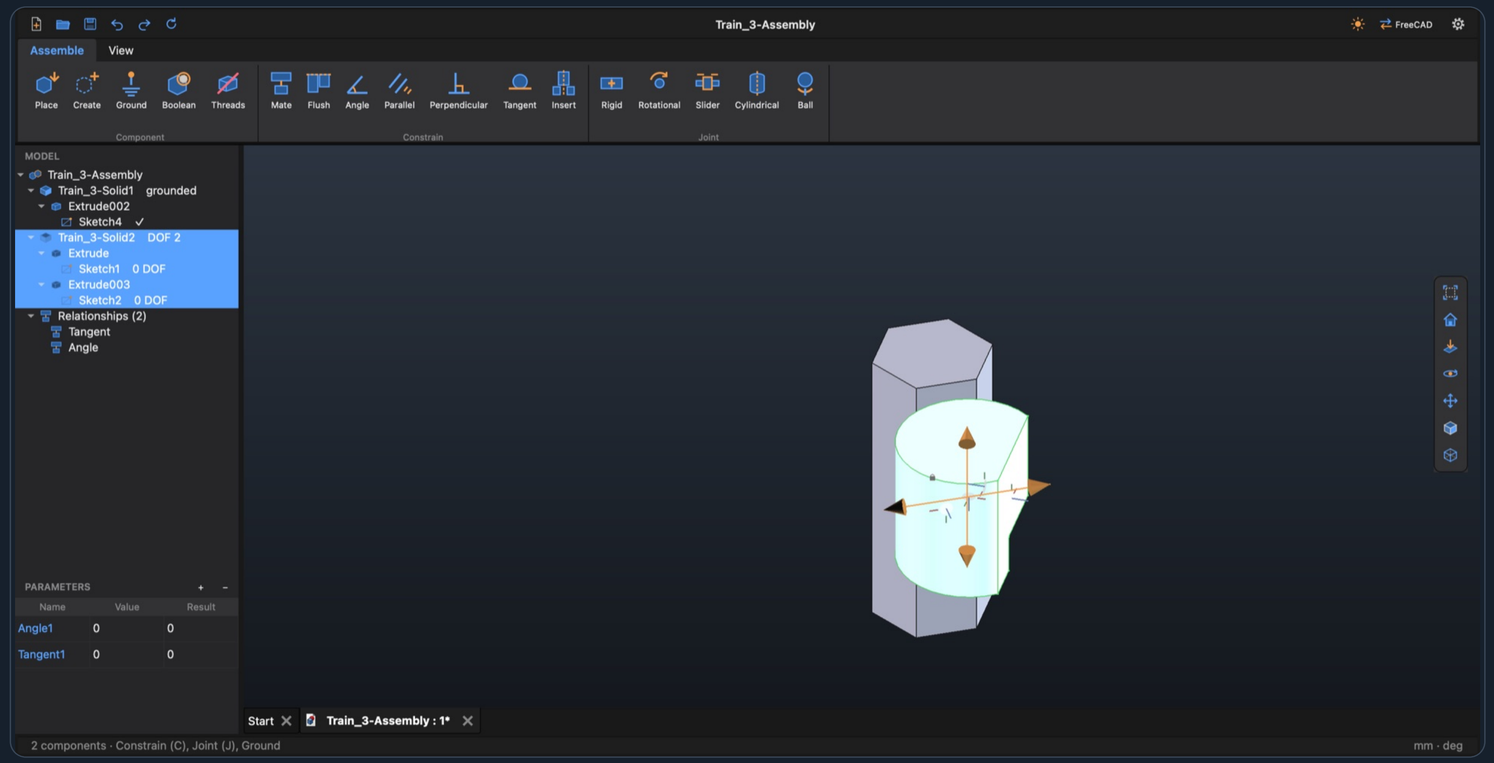
Flip turns the part over in the preview.

## RELATIONSHIPS

Double-click a row to edit it. Right-click: Edit, Delete, Suppress, Rename.

ASSEMBLIES

# Moving parts



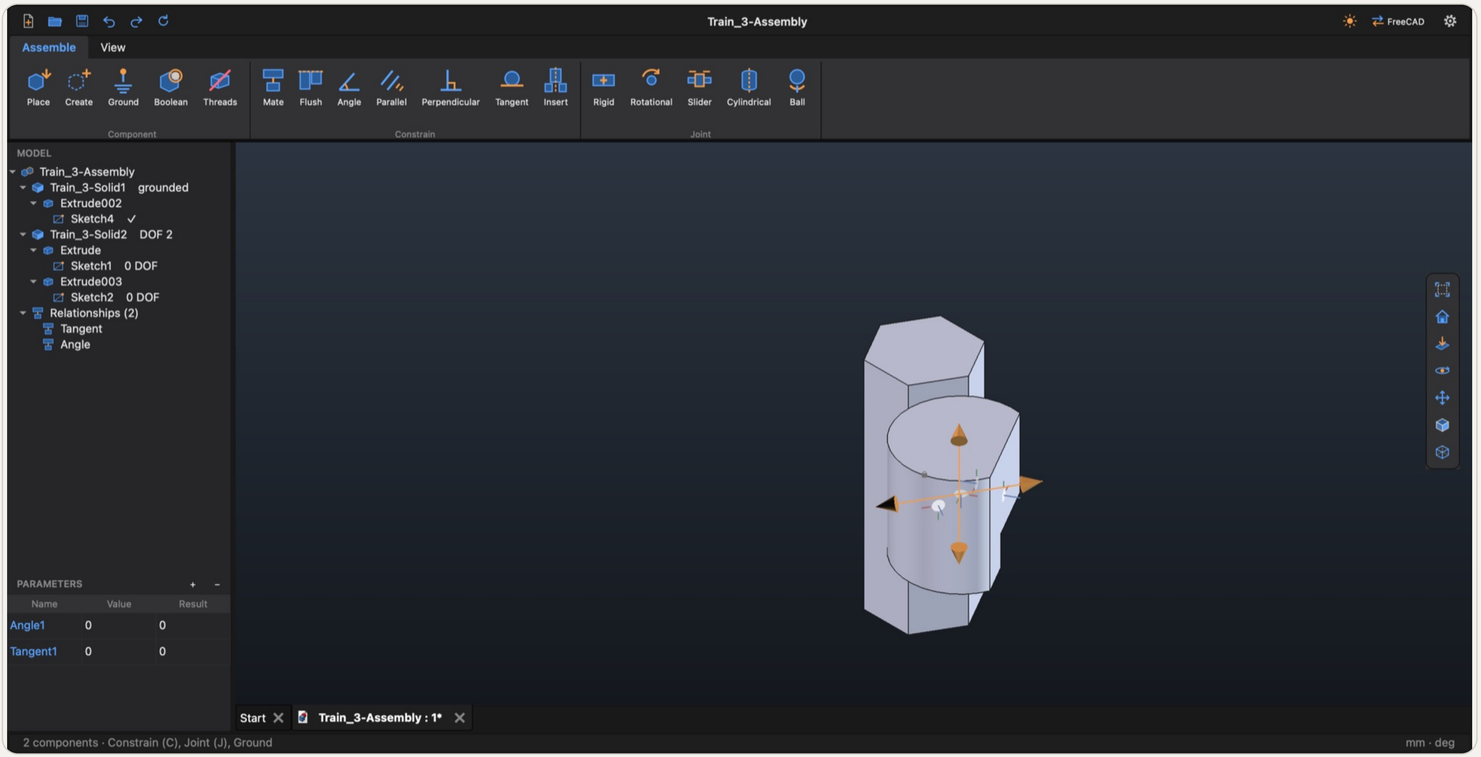

Drag a free part, or click it to get its gizmo: arrows slide, arcs turn.

Only along the part's own degrees of freedom. The other parts stay put.

One drag is one undo step.

ASSEMBLIES

# Degrees of freedom



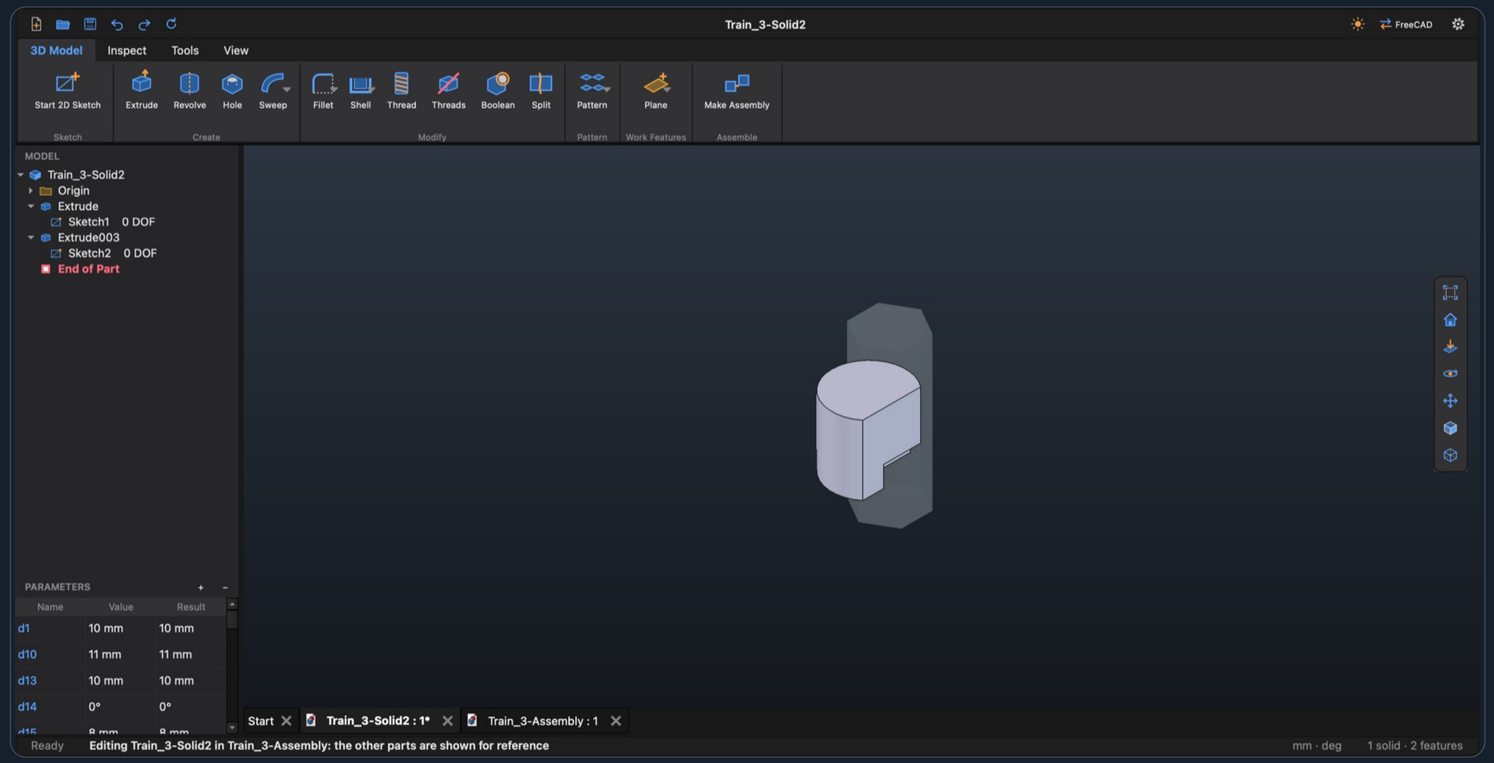

Each part's row shows Grounded or DOF n: the ways it can still move.

View › Degrees of Freedom draws Inventor's DOF symbols on the parts.

With it off, the part under the pointer shows its symbol as a tip.

ASSEMBLIES

# Edit a part in context



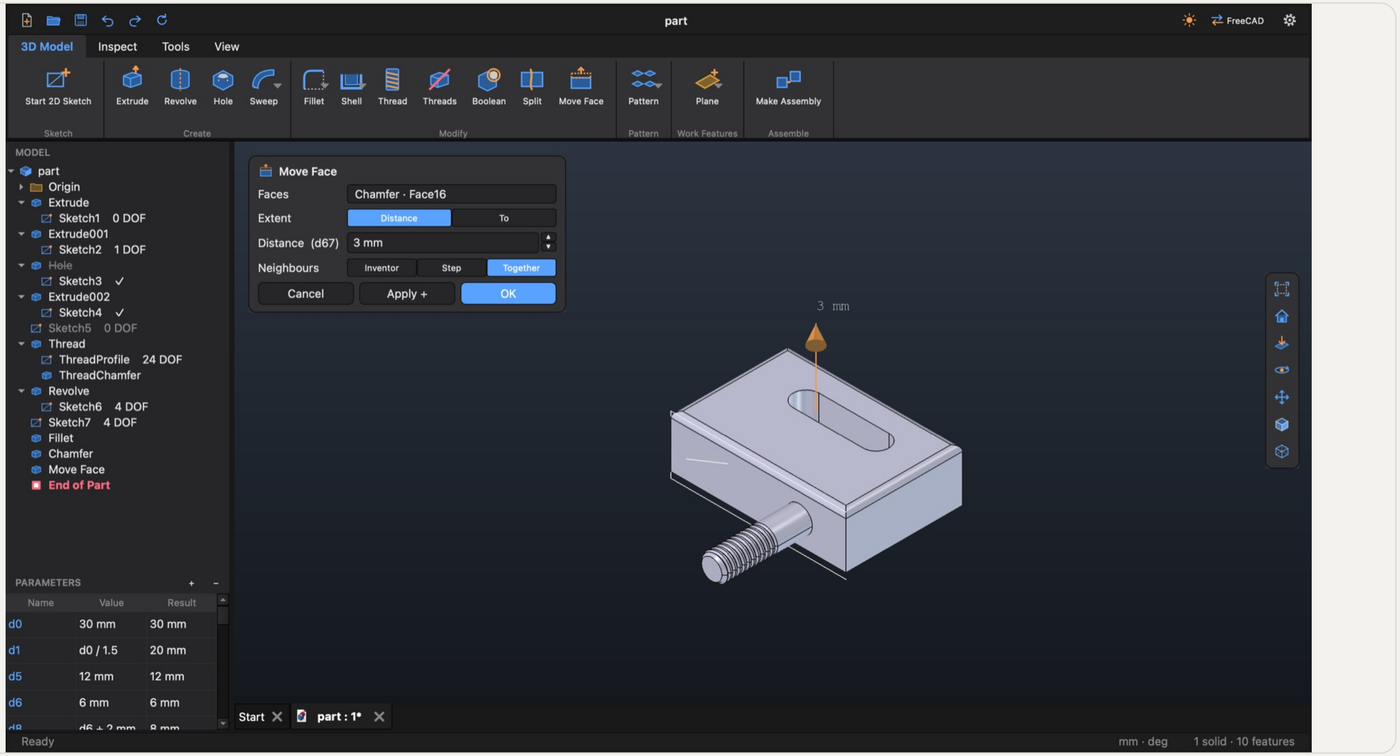

Edit opens the part in its tab with the other parts semi-transparent around it.

View › Assembly Context turns them off and on.

Back in the assembly, InCad asks before updating the part.

DIRECT EDITING

# Move Face



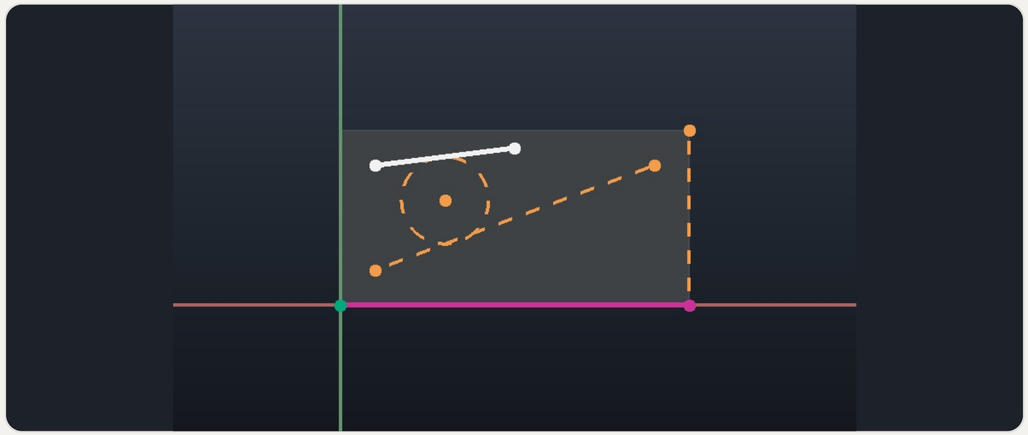

Pick flat faces and push or pull them: type a distance, or drag the arrow and read the value at its tip.

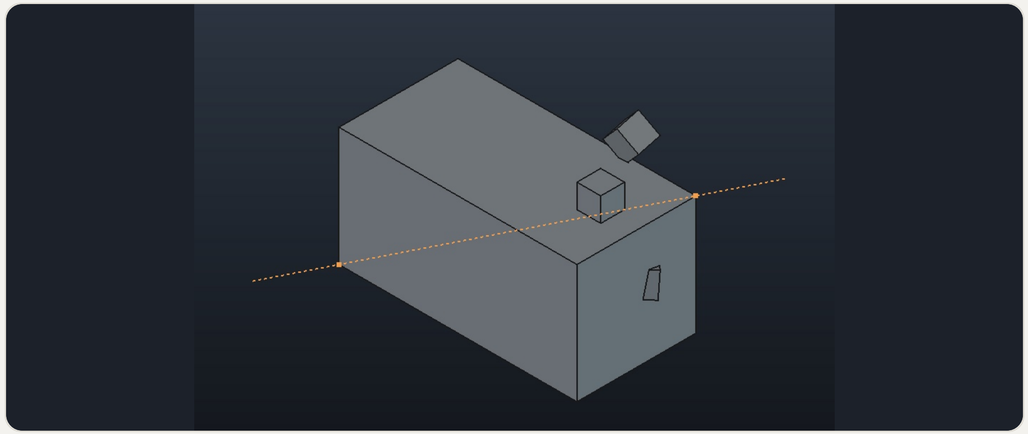
Neighbours: Inventor (chamfers keep their angle), Step, or Together (the face moves with its chamfers and fillets).

To a parallel face, plane, edge or point. A step in the history: edit, suppress, delete. Finish move on top of the right-click menu.

NEW IN 0.4

# Sketching and modelling, sharper





Offset: one click takes the whole shape; sharp corners, a line gives a parallel line, a circle or ellipse a ring.

Concentric constraint; construction is orange and dashed; the part is darker while you sketch.

Fillet / Chamfer: a click on a face takes all its edges, on an edge its smooth chain.

Circular Pattern around any axis, edge, round face, or two clicked points.

Move Face on round walls: a cylinder or an extruded ellipse moves outward or inward.

BUILD STATUS

# Early test version 0.3

| What | Status |
|---|---|
| Parts | Tested, waiting for your feedback |
| Assemblies | Ready to test (new in 0.2) |
| Move Face (direct editing) | Ready to test (new in 0.3) |
| macOS, Apple Silicon | Supported |
| FreeCAD | 1.1, tested with 1.1.3 |
| Windows, Linux, Intel Macs | Later, if there is demand |

Save often and try it on copies of important files

INSTALL

# Four steps with the Addon Manager

1. Unzip the archive and copy the path of the InCad folder: Option+Cmd+C
2. Settings › Addon Manager › Custom repositories › “+”, paste the path
3. Tools › Addon Manager › InCad › Install
4. Restart FreeCAD and choose the InCad workbench

**To start FreeCAD in InCad:** Settings › Workbenches › Available Workbenches › Startup › Default workbench: InCad

Details and the manual install: README.txt in the archive. The full manual: InCad-manual.pdf

HOW TO HELP

# Found a bug? Send four things

- **Steps:** what you did and what you expected
- **Log:** the file ~/Library/Logs/InCad.log
- **Model:** the .FCStd file, if you can share it
- **Versions:** build name, macOS and FreeCAD; after a crash, macOS's report too

# Try it and tell us what's wrong

Every report brings InCad closer to the CAD you are used to

InCad 0.3 · test build for macOS on Apple Silicon · FreeCAD 1.1